

Estradiol-17 β modifies the induction of spermidine/spermine N¹-acetyltransferase activity in the liver of lipopolysaccharide-treated mice

Sugimoto H, Matsuzaki Y, Yuasa K, Kaneko M, Yamada S, Hamana K, Matsuzaki S. Estradiol-17 β modifies the induction of spermidine/spermine N¹-acetyltransferase activity in the liver of lipopolysaccharide-treated mice. *Liver* 1992; 12: 147–151.

Abstract: In an attempt to elucidate the effects of estrogen on polyamine metabolism in lipopolysaccharide (LPS)-treated mice, we assayed polyamine content and the activity of spermidine/spermine N¹-acetyltransferase (SAT) and ornithine decarboxylase (ODC) in some organs. LPS elevated N¹-acetylspermidine levels in the liver and lung and putrescine levels in the liver, lung and spleen. LPS increased the activity of ODC at 6 h and that of SAT at 12 h in the liver. When estradiol-17 β was simultaneously administered with LPS, the maximum increase in hepatic N¹-acetylspermidine levels was found 6 h earlier than in the LPS control. Likewise, the peak of the hepatic SAT activity after LPS-treatment was observed 6 h earlier in the estradiol-17 β -treated mice than in the LPS control. No such effect of estradiol-17 β was found in the lung and spleen. The LPS-induced ODC activity was not affected by estradiol-17 β in the liver, lung or spleen. Estrone and 16 β -ethylestradiol (an anti-estrogen) were also effective in enhancing the LPS-induced elevation of N¹-acetyl-spermidine and putrescine in the liver, while both diethylstilbestrol, which has a potent estrogenic activity without steroid structure and estradiol-17 α (a non-estrogenic isomer of estradiol-17 β) were without effect. Tamoxifen (an estrogen receptor antagonist) did not suppress the estrogen-induced increase in hepatic N¹-acetylspermidine levels.

**Hiroyuki Sugimoto¹,
Yutaka Matsuzaki², Kojiro Yuasa²,
Mieko Kaneko², Shoji Yamada²,
Koel Hamana³ and
Shigeru Matsuzaki¹**

¹Department of Physiology, Institute of Endocrinology, ²First Department of Internal Medicine, School of Medicine, and ³College of Medical Care and Technology, Gunma University, Gunma Japan

Key words: estrogens – lipopolysaccharide – ornithine decarboxylase – polyamines – spermidine/spermine N¹-acetyltransferase

Hiroyuki Sugimoto M.D., Ph.D. Department of Biochemistry, Gunma University School of Medicine, 3-39-15 Showa-Machi, Maebashi, Japan 371

Received 5 November 1991, accepted for publication 13 March 1992

Putrescine, spermidine and spermine are the major polyamines which are distributed in normal animal tissues. Putrescine is formed from ornithine by ornithine decarboxylase (ODC) and putrescine is converted to spermidine, and spermidine to spermine by aminopropyltransferases using decarboxylated S-adenosylmethionine (1). Previously we showed that lipopolysaccharide (LPS) increased the levels of both putrescine and N¹-acetylspermidine in mouse liver (2). The elevation of hepatic levels of these polyamines is found to be one of the earliest events that occur after treatment with the hepatotoxic substances. This elevation was suppressed by the administration of free-radical scavengers (3). These findings seem to show that hepatotoxic compounds induce marked changes in the activity of hepatic spermidine/spermine N¹-acetyltransferase (SAT) which produces acetylpolyamines from spermidine and spermine, and of ODC which may regulate these polyamine levels.

Glucocorticoids and estrogens have immunosuppressive effects in some tissues (4, 5). For example, estrogens decrease the production of interleukin-2 and the proliferation of antigen-sensitized T lymphocytes after treatment with *Listeria monocytogenes* (5). On the other hand, estrogens stimulate lymphoid regeneration after irradiation (6). It is not clear that estrogens attenuate the toxicity of LPS-treated mice (5, 6).

Polyamines, especially putrescine, are known to act as anti-inflammatory agents in rats (7). Our previous results indicated that glucocorticoids suppress the LPS-induced elevation of putrescine and N¹-acetylspermidine in ddY mouse liver, while some estrogens enhance the elevation of these polyamines (8). However, ddY mice showed some confusing results in response to the treatment with estrogens. These mice showed rather great individual variations in polyamine levels and enzyme activities. The aim of the present study is to elucidate

which enzyme activities, ODC or SAT, are responsible for the observed changes in polyamine levels in the liver, lung and spleen of LPS-treated BALB c mice, which showed smaller variations than ddY mice.

Material and methods

Animals and treatments

Seven-week-old male BALB c mice were kept in standard plastic cages and allowed free access to a normal diet and water before experiments. All animals were given only water for the last 8 h before sacrifice.

LPS (*Escherichia coli* (055:B5), origin DIFCO Lab., U.S.A.) (30 mg/kg) dissolved in saline solution (0.2 ml) was injected simultaneously with either estradiol-17 β , estradiol-17 α , estrone, diethylstilbestrol (from Sigma, U.S.A.) or 16 β -ethylestradiol (from Takeda Pharmaceutical Co. Ltd., Japan), dissolved in olive oil (0.2 ml), and the animals were killed 3, 4, 6, 12 and 18 h thereafter. Steroid hormones were injected i.p. at a dose of 50 μ mol/kg. Tamoxifen (from ICI Pharmaceutical Co. Ltd., U.S.A.) at a dose of 150 μ mol/kg (56 mg/kg) was injected (i.p.) 30 min before estradiol-17 β treatment.

Assay of polyamines

Mouse livers weighing 100 mg were homogenized in 1 ml of cold 5% perchloric acid (HClO₄) and then centrifuged at 2000 rpm for 10 min at 4°C. Polyamines in HClO₄ extracts were analysed by HPLC on a column (4.6 $\times$ 80 mm) of cation-exchange resin (Kyowa Seimitsu, 62210FK) as described previously (2). N¹-acetylspermidine and N⁸-acetylspermidine were distinguished from each other by using deacetylases and acid hydrolysis (2). DNA in the pellets was extracted in 1 ml of 5% HClO₄ at 90°C for 15 min and assayed according to Burton (9). The polyamine concentrations were expressed as nmol/mg DNA.

Assay of ODC

The activity of ODC was measured as reported previously (10). Briefly, tissues were homogenized with a Phycotron (microhomogenizer) in three volumes of ice-cold 50 mmol/l sodium phosphate buffer (pH 7.2) containing 5 mmol/l dithiothreitol. The whole homogenates were centrifuged at 10 000 $\times$ g for 20 min to obtain post-mitochondrial fractions. The reaction mixture in a total volume of 200 μ l contained 50 mmol/l Na phosphate buffer (pH 7.2), 5 mmol/l dithiothreitol, 1 mmol/l aminoguanidine, 0.5 mmol/l ornithine, 0.05

mmol/l pyridoxal 5-phosphate, 0.1 mmol/l EDTA and 100 μ l of the tissue supernatant. The incubation was carried out at 37°C for 3 h. The reaction was linear with respect to incubation time for at least 3 h. The reaction was stopped by adding 10 μ l of 6 mmol/l perchloric acid. To this mixture 190 μ l of 0.35 mmol/l citrate buffer (pH 5.49) containing 0.5 mmol/l KCl was added. After centrifugation, the supernatants were filtered through microfilters with 0.45 μ m pores. To measure the increase in putrescine during 3 h incubation, zero time controls (without incubation) were analyzed at the same time. Putrescine was determined by HPLC as described previously (2). The ODC activities were expressed as putrescine produced per g wet weight per hour.

Assay of SAT

The activity of SAT was measured according to Matsui et al. (11). Briefly, tissues were homogenized in three volumes of 0.25 mmol/l sucrose-Na phosphate buffer (pH 7.5) containing 25 mmol/l KCl and 5 mmol/l MgCl₂ and centrifuged as described above. The reaction mixture in a total volume of 150 μ l, contained 50 mmol/l Tris, 30 mmol/l spermidine, 7.5 μ l of C¹⁴-acetyl-CoA (20 μ mol/l acetyl-CoA, 220 000 cpm) and 75 μ l of the supernatant. The incubation was carried out at 30°C for 10 min. The reaction was stopped by adding 30 μ l of NH₂Cl and boiled. Sixty μ l of the supernatant was applied on the ion-exchange paper (P 81, Whatman, diameter: 2.5 cm) and the papers were washed with 0.1 mmol/l NH₃ three times. The radioactivity on the papers was measured in a toluene scintillator with a Beckman LS7000 liquid scintillation spectrometer. The increase in SAT activity was expressed as a multiple of the control value.

Statistical analysis

All the values were expressed as mean $\pm$ SEM. Group means between the LPS-treated group, and LPS+steroid hormone-treated groups were compared by Student's *t*-test or the Cochran-Cox test after analysis of variance to determine the significance of difference between individual means. In all cases, a probability of error of <0.05 was selected as the criterion for statistical significance.

Results

Effect of estradiol-17 β on polyamine contents in LPS-treated mouse liver, lung and spleen

After treatment with LPS, putrescine increased at 3, 6 and 12 h in the liver, lung and spleen, while N¹-acetylspermidine increased only in the liver and

Table 1. Effects of estradiol-17 β on polyamine levels in liver, lung and spleen of LPS-treated mice.

	Polyamines (nmol/mg DNA)				
	N	PUT	SPD	SPM	Acetyl-SPD
<i>Liver</i>					
Control	3	7.2±1.2	729±30	1153±25	N.D.
3 h					
LPS	6	46.9±5.8 ^b	1161±79	1792±57	12.9±1.8 ^c
LPS+estradiol-17β	6	49.4±3.7 ^c	1254±63	1786±82	15.1±1.5 ^c
Estradiol-17β	3	11.1±0.3	1044±43	1463±46	N.D.
6 h					
LPS	6	105.1±10.5 ^c	1013±84	1501±101	27.9±3.0 ^c
LPS+estradiol-17β	6	132.4±6.6 ^c	1092±33	1678±35	59.1±9.6 ^{b,d}
Estradiol-17β	3	28.6±5.6	861±117	1555±151	N.D.
12 h					
LPS	3	96.7±6.2 ^c	694±24	1201±69	66.5±9.5 ^a
LPS+estradiol-17β	3	157.5±32.8 ^a	739±141	1100±98	43.8±10.1 ^a
Estradiol-17β	3	11.7±3.1	1030±158	1299±83	1.7±0.9
<i>Lung</i>					
Control	3	4.0±0.3	98.2±6.5	84.5±6.3	0.8±0.3
6 h					
LPS	6	19.6±1.3 ^c	117.8±5.8	91.7±4.5	3.4±0.3 ^b
LPS+estradiol-17β	6	16.4±0.8 ^c	111.1±3.8	86.2±2.6	3.4±0.3 ^b
Estradiol-17β	3	4.3±0.4	88.4±3.0	74.9±1.5	0.6±0.3
<i>Spleen</i>					
Control	3	6.3±0.2	261.4±21.5	243.3±23.7	1.3±0.2
6 h					
LPS	6	20.1±1.2 ^c	258.7±15.2	224.4±13.8	0.9±0.1
LPS+estradiol-17β	6	15.8±0.7 ^c	202.3±11.2	168.2±8.9	0.7±0.2
Estradiol-17β	3	9.2±1.0	299.5±8.9	261.9±4.4	0.9±0.5

PUT=putrescine; SPD=spermidine; SPM=spermine; Acetyl-SPD=N¹-acetylspermidine; N=number of determinations; N.D.=not detected. ^a P < 0.05, ^b P < 0.01 and ^c P < 0.001 compared with control groups. ^d P < 0.05 compared with LPS-treated group.

lung. The peak increase in N¹-acetylspermidine levels in the liver of LPS-treated mouse was found 6 h earlier in estradiol-17 β -treated mice than in the control. This effect was not detected in the lung and spleen (Table 1). The hepatic polyamine levels with or without LPS and estradiol-17 β treatments in BALB c mice used in this study show essentially the same changes in the hepatic polyamine levels in ddY mice as reported previously (2, 8). However, the statistical analysis showed smaller variation in polyamine content in a smaller number of BALB c mice than in ddY mice.

Effect of estradiol-17 β on the hepatic activity of ODC and SAT in LPS-treated mice

Estradiol-17 β had no significant effect on ODC activity in LPS-treated mouse liver, as shown in Fig. 1. The peak of the SAT activity in LPS-treated mice was observed 6 h earlier in estradiol-17 β -treated mice than in the LPS control (Fig. 1).

Effect of various estrogens and tamoxifen on LPS-induced increase in polyamine contents in mouse liver

Estradiol-17 β , estrone and 16 β -ethylestradiol enhanced the hepatic elevation of putrescine and N¹-acetylspermidine in LPS-treated mice at 4 h, while diethylstilbestrol and estradiol-17 α were without effect. Tamoxifen did not suppress the effect of estradiol-17 β (Table 2).

Discussion

The results of our present study have shown that estrogens modify the LPS-induced changes in polyamine metabolism in rat liver. The maximum elevation of hepatic N¹-acetylspermidine levels in LPS-treated mice at 12 h coincided with the peak of the activity of SAT. After the treatment of LPS-treated mice with estradiol-17 β , the peak activity of hepatic SAT and the peak of N¹-acetylspermidine were also observed at the same time and found 6 h earlier than the control. The levels of putrescine in

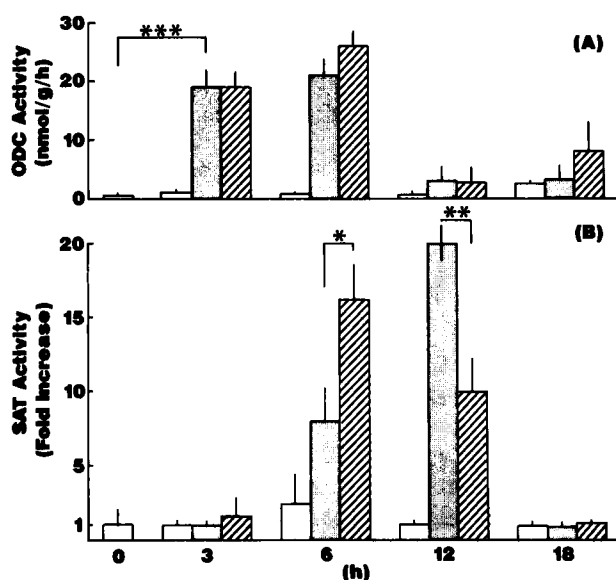


Fig. 1. Effects of estradiol-17 β on the ODC activity (A) and the SAT activity (B) in liver of LPS-treated mice. The enzyme activities were determined 0, 3, 6, 12 and 18 h after treatment with LPS (gray bars) or LPS+estradiol-17 β (hatched bars). Open bars represent control groups at 0 h and estradiol-17 β -treated groups at 3, 6, 12 and 18 h. * $P < 0.05$, ** $P < 0.02$ and *** $P < 0.01$ compared with LPS-treated and control groups, respectively. Six animals were used in each group. ODC = ornithine decarboxylase; SAT = spermidine/spermine N¹-acetyltransferase.

LPS-treated mouse liver at 4 h were lower than that in LPS-treated mouse liver with estradiol-17 β . The hepatic ODC activity of LPS-treated mice was not affected by estradiol-17 β . These results show that, after treatment with LPS, the increase in the activity of ODC and SAT preceded that in the

hepatic levels of putrescine and N¹-acetylspermidine. After simultaneous stimulation with estradiol-17 β and LPS, the increase in hepatic SAT activity was observed several hours earlier than in the LPS control but that of ODC was the same.

Estradiol-17 β , estrone and 16 β -ethylestradiol (an anti-estrogen) enhanced the polyamine elevation after LPS-treatment. However, diethylstilbestrol which is known to have a potent estrogenic activity without steroid structure and estradiol-17 α (a non-estrogenic isomer of estradiol-17 β), as well as 16 β -hydroxyestrone, 2-hydroxyestrone, 16 β -hydroxyestradiol, 2-hydroxyestradiol, progesterone and testosterone, as reported previously (8), have no such effect.

The sensitivity of BALB c mouse to some steroid hormones in these experiments was slightly different from that of ddY mouse in our previous report (8). N¹-acetylspermidine in the liver of ddY mice treated with LPS and estradiol-17 α significantly increased compared with LPS treatment (8). However, this effect of estradiol-17 β was not detected in the liver of BALB c mice as shown in this study and no significant effect of estradiol-17 α to the LPS-treated ddY mice was detected by extensive re-examination.

These results indicate that the effects of steroids on the enhancement of LPS-induced increase in putrescine and N¹-acetylspermidine require estratriene structure containing a hydroxy group at C-3 and 17 β -hydroxy or keto group at C-17. Tamoxifen, one of the estrogen receptor antagonists, did not cancel the effect of estradiol-17 β in LPS-induced increase in putrescine and N¹-acetylspermidine. The effect of estradiol-17 β on the enhancement of the elevation in LPS-treated mouse was not

Table 2. Effects of estradiol-17 β , tamoxifen, estrone, 16 β -ethylestradiol, diethylstilbestrol, estradiol-17 α on polyamine levels in liver of LPS-treated mice at 4 h.

	N	Polyamines (nmol/mg DNA)			
		PUT	SPD	SPM	Acetyl-SPD
Control	3	5.1 $\pm$ 1.0	491 $\pm$ 28	823 $\pm$ 7	N.D.
LPS	6	42.4 $\pm$ 3.3 ^a	509 $\pm$ 22	802 $\pm$ 21	15.2 $\pm$ 2.0 ^a
LPS+estradiol-17 β	6	58.4 $\pm$ 5.2 ^b	586 $\pm$ 36	897 $\pm$ 37	31.0 $\pm$ 5.5 ^b
LPS+estradiol-17 β +tamoxifen	7	58.8 $\pm$ 4.9 ^b	577 $\pm$ 20	925 $\pm$ 34	36.8 $\pm$ 6.2 ^b
LPS+estrone	6	60.1 $\pm$ 5.1 ^b	563 $\pm$ 30	887 $\pm$ 46	35.9 $\pm$ 4.8 ^c
LPS+16 β -ethylestradiol	5	64.8 $\pm$ 11.9	458 $\pm$ 47	802 $\pm$ 70	33.8 $\pm$ 6.3 ^b
LPS+diethylstilbestrol	6	38.1 $\pm$ 3.8	473 $\pm$ 42	770 $\pm$ 67	13.6 $\pm$ 2.3
LPS+estradiol-17 α	5	44.5 $\pm$ 3.9	447 $\pm$ 20	682 $\pm$ 22	9.4 $\pm$ 1.2
Estradiol-17 β	9	8.4 $\pm$ 1.4	447 $\pm$ 23	806 $\pm$ 40	N.D.
Tamoxifen	3	6.2 $\pm$ 1.2	528 $\pm$ 96	794 $\pm$ 50	2.3 $\pm$ 1.2
Estrone	8	4.6 $\pm$ 0.3	525 $\pm$ 26	839 $\pm$ 33	N.D.
16 β -ethylestradiol	4	4.9 $\pm$ 0.5	383 $\pm$ 48	681 $\pm$ 61	N.D.
Diethylstilbestrol	3	8.3 $\pm$ 1.3	604 $\pm$ 88	986 $\pm$ 163	N.D.
Estradiol-17 α	3	17.4 $\pm$ 4.7	884 $\pm$ 74	1484 $\pm$ 84	N.D.

PUT=putrescine; SPD=spermidine; SPM=spermine; Acetyl-SPD=N¹-acetylspermidine; N=number of determinations; N.D.=not detected. ^a $P < 0.001$ compared with control. ^b $P < 0.05$, ^c $P < 0.01$ compared with LPS-treated group.

detected in lung and spleen. These results suggest that estrogens enhance the SAT activity to elevate N¹-acetylspermidine level only in LPS-treated mouse liver by a certain estrogenic steroid structure-dependent mechanism which is not directly mediated by the steroid hormone receptor system.

Both putrescine and spermidine are known to act as anti-inflammatory agents in rats (7). Polyamines, especially putrescine, scavenge superoxide anion produced by rat polymorphonuclear leukocytes stimulated with zymosan *in vitro* (12). Putrescine also decreases the amount of acid phosphatase liberated from lysosomes incubated in the presence of xanthine oxidase and hypoxanthine (13). Estrogens sometimes stimulate cell growth in certain cells. For example, estrogens enhance the activity of macrophages (14). Estrogens also enhance the lymphoid regeneration and immune response in irradiated and marrow-reconstituted mice (6). Hepatocellular hyperplasia is observed after estrogen administration in CCl₄-treated mouse (15). After partial hepatectomy or administration of growth hormone, the activity of SAT increases in rat and mouse liver (16). The enhanced elevation of putrescine by estrogens through the activation of SAT in the early stage of endotoxic shock may attenuate the damage of hepatocytes, and then hyperplasia of the hepatocytes may take place.

References

1. PEGG A E, McCANN P P. Polyamine metabolism and function. *Am J Physiol* 1982; **243**: C212-C221.
2. SUGIMOTO H, YAMADA S, ARAI T, KOBAYASHI S, HAMANA K, MATSUZAKI S. Elevation of acetylpolyamine levels in mouse tissues, serum and urine after treatment with radical-producing drugs and lipopolysaccharide. *Hepatology* 1988; **8**: 267-271.
3. SUGIMOTO H, MATSUZAKI S, HAMANA K, YAMADA S, KOBAYASHI S. α -tocopherol and superoxide dismutase suppress and diethyldithiocarbamate and phorone enhance the lipopolysaccharide-induced increase in N¹-acetylspermidine concentrations in mouse liver. *Circ Shock* 1991; **33**: 171-177.
4. OLSON N C, BROWN T T, ANDERSON D L. Dexamethasone and indomethacin modify endotoxin-induced respiratory failure in pigs. *J Appl Physiol* 1985; **58**: 274-284.
5. PUNG O J, TUCKER A N, VORE S J, LUSTER M I. Influence of estrogen on host resistance: increased susceptibility of mice to *Listeria monocytogenes* correlates with depressed production of interleukin 2. *Infect Immun* 1985; **50**: 91-96.
6. KOTANI M, FUJII H, TSUCHIYA H. Effects of estrogen on the lymphoid regeneration and immune response in irradiated and marrow-reconstituted mice. *Acta Anat* 1979; **105**: 298-308.
7. BIRD J, HIRDIR S M, LEWIS D A. Putrescine - a potent endogenous anti-inflammatory substance in inflammatory exudates. *Agents Actions* 1983; **13**: 342-347.
8. SUGIMOTO H, HAMANA K, MATSUZAKI S, ARAI T, YAMADA S. Glucocorticoids suppress and estrogens enhance the lipopolysaccharide-induced increase in putrescine and N¹-acetylspermidine in mouse liver. *J Steroid Biochem* 1991; **38**: 781-786.
9. BURTON K. A study of the conditions and mechanism of the diphenylamine reaction for the colorimetric estimation of deoxyribonucleic acid. *Biochem J* 1956; **62**: 315-323.
10. ARAKI M, NAKAMURA M, TAKENOSHITA S, SHODA H, NAGAMACHI Y, MATSUZAKI S. Effects of dexamethasone on the activity of histidine decarboxylase, ornithine decarboxylase, and DOPA decarboxylase in rat oxyntic mucosa. *Can J Physiol Pharmacol* 1991; **69**: 37-42.
11. MATSUI I, WIEGAND L, PEGG A E. Properties of spermidine N-acetyltransferase from livers of rats treated with carbon tetrachloride and its role in the conversion of spermidine into putrescine. *J Biol Chem* 1981; **256**: 2454-2459.
12. KAFY A M L, HAIGH C G, LEWIS D A. *In vitro* interactions between endogenous polyamines and superoxide anion. *Agents Actions* 1986; **18**: 555-559.
13. KAFY A M L, LEWIS D A. Antioxidant effects of exogenous polyamines in damage of lysosomes inflicted by xanthine oxidase or stimulated polymorphonuclear leucocytes. *Agents Actions* 1988; **24**: 145-151.
14. VERNON-ROBERTS B. The effects of steroid hormones on macrophage activity. *Int Rev Cytol* 1969; **25**: 131-159.
15. OGIHARA Y, SUGIMOTO M. Influence of estradiol and testosterone on carbon tetrachloride-induced liver damage in the rat. *Jpn J Gastroent* 1991; **88**: 1319-1327.
16. MATSUI I, PEGG A E. Effect of thioacetamide, growth hormone or partial hepatectomy on spermidine acetylase activity of rat liver cytosol. *Biochim Biophys Acta* 1980; **633**: 87-94.